Coverage Diversity in Mega Satellite Constellations: A Stochastic Geometry Approach

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***Abstract*—To keep up with the continuously growing coverage demands and attain true global coverage, the deployment of multi-layered low Earth orbit satellite constellations is necessary. Next-generation mega satellite constellations are expected to rely on inter-satellite links to relay information, which will enable fast and reliable communications between the different satellite nodes in free-space and facilitate the utilization of coverage diversity modes that can further enhance the quality-of-service provided in the network. However, materializing these high performing systems is challenging due to the complexity of the network architecture which may require long and complex simulation processes during design. In this article, we develop theoretical modeling for the probability of coverage for various diversity modes in mega satellite constellations by leveraging tools from stochastic geometry. We first develop analytical models for conventional single-shell networks and then extend these models to incorporate multi-shell networks. The analysis is validated using Monte-Carlo simulations which show a close fit to the analytical models derived. Moreover, the analytical models provide a performance baseline that is comparable to practical networks that rely on regular network architectures such as SpaceX's Starlink. This allows network operators to devise expansion strategies to cater for expanding demands and gain insights into the performance of the network as more shells are introduced into the network.**

***Index Terms*—Stochastic geometry, mega satellite constellation, low earth orbit, cell-free massive MIMO.**

I. Introduction

ADVANCES in telecommunications are yet to be matched by network capabilities to attain global coverage and low latency communications. Owing to its inherent geographic and economic limitations, the current terrestrial infrastructure remains lacking in its ability to facilitate true global coverage [1], [2]. Therefore, to compensate for the shortcomings of the current terrestrial networks, next-generation wireless networks are envisioned to rely on multi-layered dimensions, predicated on combinations of terrestrial and non-terrestrial network architectures [3], [4], [5]. Over the past few years, the space industry has witnessed revolutionary leaps in satellite technology, transitioning from the few bulky geostationary satellites to interconnected mega low-Earth orbit (LEO) constellations [6]. With their exceptionally high coverage capacity, spectral energy efficiency, and compatibility with existing terrestrial infrastructure [2], satellite constellations are integral for the path to materializing a holistic 6G ecosystem. As a result, network providers are transitioning towards LEO communications with the deployment of mega satellite constellations, such as SpaceX's Starlink, OneWeb, and Amazon's Kuiper. This transition is further fueled by the advancement of multi-antenna multi-beam transceivers, multi-shell deployments, and inter-satellite links (ISLs) which serve as forward-looking enablers for next-generation satellite communications [7], [8].

A. Diversity in Satellite Communications

Despite the difficulty of unreliable communication channels, there have been efforts to integrate such transmission mediums in the earlier and current generations of cellular networks using well-established wireless techniques, namely multiple-input and multiple-output (MIMO) [9]. The core concept of MIMO techniques is strategically exploiting multiple antennas at the transmitting and receiving ends of the wireless system to create a higher channel rank with uncorrelated data streams. This can result in the increase of spatial multiplexing gain and/or a significantly improved signal-to-interference-plus-noise ratio (SINR), popularly referred to as diversity gain [10]. However, the heavy utilization of geostationary satellites limited the practicality of implementing MIMO systems in satellite communications. Nevertheless, with the utilization of ISLs in LEO satellite constellations, enhancement can now be achieved by exploiting the spatial diversity of the satellites. The upsurge in implementation for MIMO configuration over satellite networks for gigabit wireless systems is exploited in the literature [11]. This is achieved through the adequate exploitation of all types of MIMO gains, where the potential applications and impacts on the space industry have been extensively discussed [12]. For instance, the mobile satellite's dual-polarization per broadcasting transmission has been deemed the first consolidated architecture that paved the non-terrestrial paradigm for leveraging the advantages of MIMO-powered networks

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[13], [14]. Further, the enhanced performance of the dual-polarization MIMO channel was confirmed [15]. However, the benefits of dual polarization in satellite communications are diminished if the satellite operates at extremely high frequencies or if a negative dominant line-of-sight (LoS) is present in a non-scattering satellite channel across the polarization configuration [16]. Alternatively, Schwarz et al. argued that spatial multiplexing is the most effective MIMO implementation for satellite communications [17], [18], [19], [20]. Provided that practical considerations and constraints at both the satellites and ground terminals were taken into account, they highlighted that optimal antenna geometry interpretation and separation, similar to the proposal in [10], are the key factors resulting in the highest spectral efficiency, surpassing all other MIMO gains in a dominant LoS satellite channel. This has been proven successful through various prototypes, such as [21] and [22], where they demonstrated the benefits of spatial gain in satellite communications using multiple single-antenna geostationary satellites in the former setup and LEO satellites in the latter work.

B. Stochastic Geometry in Satellite Communications

Stochastic geometry tools serve as a practical mathematical framework for modeling the spatially distributed network nodes, namely LEO satellites and ground users, and their analytical performance attributes, such as coverage, rate, and SINR [23], [24], [25]. In recent literature, tractable models have been employed to investigate the optimal performance of a multi-beam satellite-based network in efficiently serving multiple terrestrial users, despite the presence of terrestrial and inter-beam interference. For instance, the model in [26] is based on the locality of the users and the captured randomness of the relevant and interfering nodes, aiming to select the best user based on the highest received SINR from a multi-beam geostationary satellite in time-varying channels. By implementing this approach, the interference is successfully reduced, and the probability of coverage for a group of ground terminals is significantly improved. Furthermore, the approach was expanded to include a fixed-altitude single spherical shell constellation of massive LEO satellites spatially distributed following the Binomial point process (BPP) model, encompassing a multi-beam satellite serving arbitrary users on Earth's surface and interfering satellites on the same shell in addition to the interference mentioned earlier.

Numerous stochastic geometry-based models have evaluated the performance of emerging ultra-dense LEO satellite constellations to fulfil futuristic network requirements [27], [28]. For instance, the downlink coverage probability of the LEO constellation network is analyzed over Rayleigh fading channel in [29] and shadowed-Rician fading channels in [30]. It is worth noting that both studies used a set of randomly placed satellites on a sphere according to the BPP model within the analytical constellation, which contrasts the satellites' deterministic distribution in the state-of-the-art Walker constellations. Besides, the statistical research in [31] precisely examined the impact of the positioning of BPP-modeled satellites in a concentric multi-sphere constellation on the distance distribution of terrestrial users and nearest neighboring satellites. In [32], a new parameter has been presented to resolve the disparity between unevenly and uniformly distributed satellites in the practical and theoretical constellations. This parameter, known as the effective number of satellites, considers the geometry of the satellite shell in the constellation at a specific location on Earth; hence, the parameter forms the basis for deriving statistical expressions in a stochastic-based inclined LEO constellation model. Furthermore, the system model in [33] has incorporated the PPP model to describe the randomness of the spatial location for deployed satellite gateways on the earth's surface as relays in conjunction with the BPP-modeled satellites and users. In contrast, the spatial PPP-model distribution for satellite locations [34], [35], has become the main aspects of the constellation's analytic framework to enhance the model's applicability. A general stochastic assessment to evaluate the impact of PPP-distributed interfering terrestrial nodes on the uplink efficiency of the constellation is provided in [36]. For accurate and actual modeling of satellite positions in mega constellations towards the envisioned 6G architecture, it is proposed to utilize a non-homogeneous Poisson point process (NPPP), as outlined in [37]. This involves adjusting the modeling intensity to accommodate varying densities across different latitudes.

C. Motivation and Contributions

Distinct from those mentioned above, this article proposes a highly effective spatial diversity method for improving connectivity through utilizing an innovative design of the cell-free massive MIMO technique with the aid of the ultra-dense multi-shell satellite communications, which is empowered by ISL technology. To this end, spatial diversity integrated into satellite communications of multiple serving access satellites simultaneously and coherently located at different shells of mega satellite constellations, has been considered as the potential solution to guarantee faster and uninterrupted relaying of information at the speed of light. The result is a seamlessly efficient satellite communications system that will drastically enhance the connectivity of different spatially uncorrelated ground users. This proposed model utilizes a tractable spatial diversity approach that emphasizes the importance of a central unit within the MIMO system. For optimal performance, the unit guarantees that the replicated transmitted information is uncorrelated and co-phased by exploiting the channel state information (CSI) for all possible independent paths [38]. The system employs post-detection diversity-combining techniques [39], to effectively combat the detrimental effects of fading and its correlation impact on transmitted signals [40]. CSI estimation is one of the most complicated and well-researched topics for various channels including the satellite and aerial channels [41]. These techniques, such as maximal-ratio combining (MRC) and equal-gain combining (EGC), combine the detected received copies with varying weights in the presence of limited-noise [42]. Moreover, previous works that utilize stochastic geometry to analyze satellite communications do not incorporate the effect of spatial diversity and rely on the nearest neighbor analysis, i.e., the nearest satellite, as the main receiver. Moreover, they only rely on analyzing a

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TABLE I
NOTATION DESCRIPTION

Notation	Description
M, N	Number of shells and total number of satellites in the constellation
$R_{\oplus}$	Earth's radius
m, n, k	Shell, satellite, ground terminal indices
N_m, R_m, α_m	Number of satellites per shell, altitude of constellation per shell, and constant per shell
$\psi_m, \psi_m^{\mathrm{o}}, \psi_m^{\max}$	Effective bandwidth, horizon beamwidth, and maximum beamwidth per shell
$\psi_{\mathrm{t}}, \psi_{\mathrm{s}}$	Ground terminal and satellite bandwidths
$\Phi, \lambda_{\mathrm{o}}, \lambda, D$	User point process, total user density, active user density, and spatial duty cycle
$\mathcal{A}_m, \mathcal{B}_m$	Satellite footprint and local dome per shells
$\mathcal{S}(\mathcal{A}), \mathcal{S}(\mathcal{B})$	Space of the satellite footprint $\mathcal{A}$ on Earth's surface and the user local dome $\mathcal{B}$ in the constellation shell
x, y	Point coordinate representation for ground terminals and satellites
$\bar{N}_m, \bar{K}_m$	Mean number of satellites per local dome per shell and mean number of ground terminals per footprint per shell
$B_{\mathrm{c}}, T_{\mathrm{c}}, B$	Coherence bandwidth, coherence time, and system bandwidth
$h_{n,k}, g_{n,k}, \varphi_{n,k}$	Channel, shadowed small-scale fading, and global-zenith angle between the nth satellite and the kth user
$l_{n,k}, \zeta_{n,k}$	Free-space-path-loss and excess path-loss between the nth satellite and the kth user
$\rho_{\mathrm{t}}, G_{\mathrm{t}}, G_{\mathrm{r}}$	Transmit power, ground user antenna gain, and satellite antenna gain
$S_{n,o}, I_n, W_{\mathrm{s}}$	Received power from target user at n^{th} satellite, interference power seen by n^{th} satellite, and average noise power at satellite receiver
$\bar{I}_m$	Average aggregate interference per shell
$\gamma, \gamma_{\mathrm{o}}$	SINR and threshold SINR

constellation that composed of one shell only. To this end, the main contributions of this paper are summarized as follows:

- Provides an analytical framework for the uplink performance of single-shell mega satellite networks for two diversity schemes; (i) dominant satellite receiver and (ii) MRC combining diversity, in terms of the coverage using tools derived from stochastic geometry. Contrary to previous related works such as [34] and [35] which solely rely on analyzing the nearest satellite, this framework exploits the diversity that can be offered by mega constellations.
- Extends the analytical framework to incorporate multi-shelled mega satellite networks for the two diversity schemes in terms of the coverage using tools derived from stochastic geometry. Contrary to previous related works such as [27] and [28], this analysis accounts for multi-shelled satellite constellations which has not been yet addressed previously in the literature. This architecture is more practical since most real satellite providers, such as Starlink, utilize multi-shells in their constellations.
- Demonstrates insights into the theoretical bounds that are attained by utilizing diversity in mega satellite networks via inter-satellite links. Results show that multi-shelled constellations can push the boundary and enable higher user traffic facilitating more reliable communications and their integral part in next-generation networks.

D. Notation

Boldface lower case and upper case symbols represent vectors and matrices respectively. Conjugate, Transpose and Hermitian (complex conjugate) transpose are denoted by $(\cdot)^*$, $(\cdot)^{\mathsf{T}}$ and $(\cdot)^{\mathsf{H}}$ respectively. The expectation and variance operators are denoted by $\mathbb{E}[\cdot]$ and $\mathrm{var}[\cdot]$ respectively. Notations $\mathbb{C}^{M\times 1}$ represent complex M-dimensional vectors and $\mathbb{C}^{M\times N}$ represents complex $M \times N$ matrices. Moreover, $\mathcal{L}_{\mathrm{X}}[\cdot]$ is the Laplace transform for a random variable X is defined as $\mathcal{L}_X(s) = \mathbb{E}[e^{-sX}]$ is evaluated at $s \in \mathbb{C}$. Moreover, Table I lists the notations and variables used.

II. System Model

A. Geometric Model

We consider a dense satellite constellation comprising M satellite shells located at different altitudes, where each shell is composed of satellites orbiting Earth. We assume that the number of satellites per shell is represented by N_m, where $m \in \{1, \ldots, M\}$ represents the index of the shell. Hence, the total number of satellites in the network, denoted with N is calculated as follows

$$N = \sum_{m=1}^{M} N_m. \tag{1}$$

Moreover, the satellites per shell are assumed to orbit Earth in circular paths (for simplicity) at a fixed altitude, denoted by R_m, above Earth's mean sea level. We also assume that the satellites are connected with each other via inter-satellite links (ISLs), i.e., ISLs can be established using optical or radio frequency (RF) links. Furthermore, we assume that a central processing unit (CPU) is deployed on one or more select satellites that have superior processing capabilities, referred to as super nodes, whereby the satellites relay the received communication for processing. While backhaul transmissions are subject to many imperfections resulting in transmission errors, we assume that the satellites are connected via error-free ISLs and only focus on the ground-to-satellite link.

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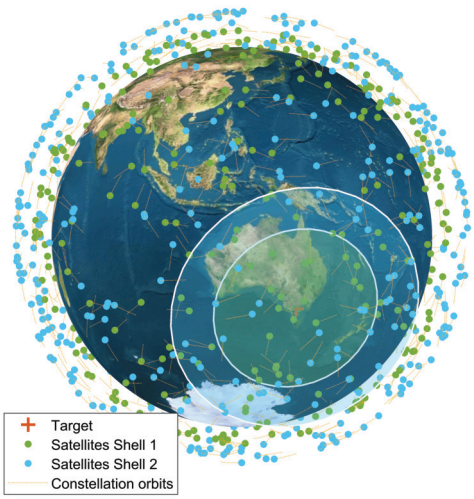

Fig. 1. Geometric representation of a two-shelled BPP satellite constellation orbiting Earth. Only the satellite that are located within the local dome per shell are reachable with respect to the target user.

To attain tractability, we assume that the satellites are uniformly distributed and are orbiting Earth in random uncorrelated circular paths, i.e., the satellite locations remain random over time. The satellite locations within a single shell at any given time snap can then be approximated using a random constellation, such as the binomial point process (BPP) [31], [33], [34], [43]. Fig. 1 depicts an example of the geometric distribution of a multi-shelled BPP orbiting Earth. While this random distribution is not attained in reality, since practical constellations follow more regular patterns, LEO satellites are typically deployed according to regularly shaped constellations such as the Walker constellations [44], [45], they have shown to provide tractable analytical lower bounds of the system performance [29], [43].

On the other hand, the ground users are assumed to also be uniformly distributed over Earth's surface according to another independent and homogeneous BPP, denoted by $\Phi \subset \mathbb{R}^3$. Furthermore, the active users are represented with a constant density, $\lambda = D\lambda_{\rm o}$ where D is the spatial duty cycle and $\lambda_{\rm o}$ is the overall user density. Moreover, the satellites in the network are assumed to jointly serve these active single antenna ground users that share the same time-frequency radio resources, i.e., a CF architecture is implemented, contrary to previous works relying on the analysis based on the nearest satellite [34].

Unlike terrestrial networks, each satellite is limited by either (i) the occlusion that occurs due to the horizon or (ii) the effective beamwidth of the antennas in the communication link [46]. As a result, the maximum Earth-centered zenith angle per shell, i.e., the zenith with respect to the Earth's center, is calculated as follows

$$\varphi_m^{\max} = \begin{cases} \operatorname{asin}\left(\dfrac{1}{\alpha_m}\sin\dfrac{\psi_m}{2}\right) - \dfrac{\psi_m}{2} & ,\psi_m < \psi_m^{\rm o} \\ \operatorname{acos}\alpha_m & ,\psi_m > \psi_m^{\rm o}, \end{cases} \tag{2}$$

where $\alpha_m = R_\oplus/(R_\oplus + R_m)$, where $R_\oplus$ is Earth's average radius, $\psi_m^{\rm o}$ is the beamwidth that meets the horizon per shell, i.e., $\psi_m^{\rm o} = 2\operatorname{asin}\alpha_m$, and ψ_m is the effective beamwidth per shell calculated as follows [46]

$$\psi_m = \min\left[\psi_{\rm s}, 2\operatorname{asin}\left(\frac{1}{\alpha_m}\sin\frac{\psi_{\rm t}}{2}\right)\right], \tag{3}$$

where $\psi_{\rm s}$ and $\psi_{\rm t}$ are the satellite and user beamwidths respectively. Note that for simplicity, we assume that the satellite and ground user beamwidths are constants, i.e., all satellites and ground users share the same beamwidth.

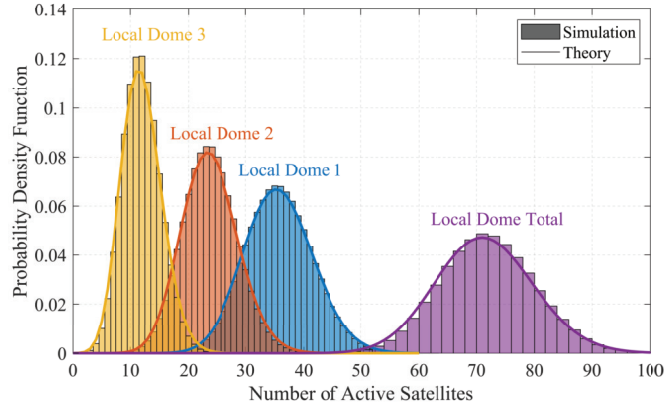

Fig. 2. Distribution of number of active satellites within a three-shell satellite constellation. The theoretical line follows PPP distribution obtained in Eq. (4).

Following the user-centric approach, each user is simultaneously served by the satellites that are within its finite local domes, where the local dome is the section of the constellation sphere within which satellites are able to connect to the ground user. Let the location of the ground users on Earth's surface be represented by $x \in \Phi$, where each user has a finite-sized local dome per shell $\mathcal{B}_m$ above it, occupying space $\mathcal{S}(\mathcal{B}_m)$ which denotes the Lebesgue measure of $\mathcal{B}_m$. Moreover, let the location of each satellite per shell be denoted by y, then the target user associates with the satellites that are within its local dome, $y \in \mathcal{B}_m$. While the number of satellites per shell and number of shells are constant, the locations of the satellites within the shell are random. As a result, the number of satellites that are within each user local dome is a random variable that can be approximated using the Poisson point process (PPP). Accordingly, the probability of there being n satellites per shell that are reachable by a given user at random is obtained as follows

$$\mathcal{P}_m^{\rm sat}(n) = \frac{\bar{N}_m^n}{n!}\exp\left(-\bar{N}_m\right), \tag{4}$$

where $\bar{N}_m$ is the average number of satellites per shell that are within the local dome and is derived as follows

$$\bar{N}_m = N_m\frac{\mathcal{S}(\mathcal{B}_m)}{4\pi(R_\oplus + R_m)^2} = \frac{N_m}{2}[1 - \cos\varphi_m^{\max}]. \tag{5}$$

Owing to the Poisson property, the probability of n satellites that serve the ground user among all the shells is obtained as

$$\mathcal{P}^{\rm sat}(n) = \frac{\left(\sum\limits_{m=1}^{M}\bar{N}_m\right)^n}{n!}\exp\left(-\sum_{m=1}^{M}\bar{N}_m\right). \tag{6}$$

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The distribution of the number of reachable satellites within a multi-shelled satellite constellation can be approximated using a Poisson distribution as depicted in Fig. 2. Similarly, each satellite can serve the ground users that are within its footprint, i.e., the section of Earth's surface that the satellite can provide coverage also dependent on the satellite altitude and the effective antenna beamwidth [43], denoted by $\mathcal{A}_m$. Thus, the average number of reachable users that are located within a satellite footprint is derived as follows

$$\bar{K}_m = \lambda \mathcal{S}(\mathcal{A}_m) = 2\pi\lambda R_{\oplus}^2[1 - \cos\varphi_m^{\max}], \tag{7}$$

where the footprint size is dependent on the satellite shell. Note that the footprint is typically extremely large and therefore a vast number of users can be served by each satellite.

B. Channel Model

In this analysis, we assume a land-mobile-system (LMS) with a frequency band in the L/S bands. Hence, we consider the effect of both the large- and small-scale fading for satellite-to-ground communications. While the majority of satellite communication occurs in vacuum, i.e., under free-space conditions, small-scale fading and shadowing are present, especially in urban environments [47]. Contrary to conventional terrestrial link analyses where only the user is assumed to be mobile, LEO satellites orbit Earth at relatively large speeds, whereby a typical LEO satellite has very short pass durations over a ground user [48]. Hence, the elevation angle of any given ground user is rapidly varying with respect to time. As a result, both the large- and small-scale fadings are varying rapidly due to the link distance by virtue of the elevation angle. Therefore, the coherence period, denoted with T_{c}, is significantly smaller in comparison to terrestrial channels.

Stochastic geometry is typically deployed to obtain the average performance and only a snapshot in time is analyzed, where it must be smaller in duration to the coherent time. Considering the geometric model presented in Sec II-A, it is reasonable to assume that the channels between the different ground users and different satellites are spatially uncorrelated, since they are assumed to be separated by multiple wavelengths [20]. Moreover, the CPU is assumed to be capable of perfectly aligning the different copies of the transmitted signals, i.e., perfect synchronization in time. Note that this assumption requires the satellites to be completely synchronized, which is extremely difficult in practice. Since the time snap considered is extremely small, we assume that there is little variation in Doppler.

Lemma 1: Let $h_{n,k}$ denote the channel vector between the kth ground user and the nth satellite in the network, then the complex baseband channel is expressed as

$$h_{n,k} = g_{n,k}\sqrt{l(\varphi_{n,k})}\exp\left(-\frac{j}{2\sqrt{l(\varphi_{n,k})}}\right), \tag{8}$$

where $l(\cdot)$ is the magnitude of the large-scale fading and g is the shadowed small-scale fading.

Proof: See Appendix A. ■

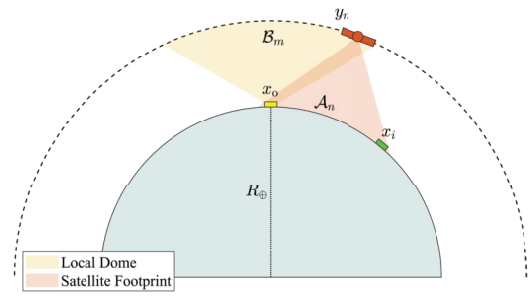

Fig. 3. Geometric representation of a typical interferer that is within the footprint of the receiver satellite.

Since the vast majority of the link is in vacuum, a big portion of the fading is mostly free-space path-loss which is dependent on the zenith angle and is expressed as

$$l(\varphi) = \frac{l_{\mathrm{o}}}{R_{\oplus}^2 + (R_{\oplus} + R_m)^2 - 2R_{\oplus}(R_{\oplus} + R_m)\cos\varphi}, \tag{9}$$

where $l_{\mathrm{o}} = (c/4\pi f_{\mathrm{c}})^2$ is the path-loss constant with f_{c} being the carrier frequency and c the speed of light. Moreover, we define the small-scale fading power as $\zeta = |g|^2$.

C. Signal Power Model

The received down-converted I/Q signal is the concatenation of the received signals from the different satellites that are within the target user's local dome. Let $x_{\mathrm{o}} \in \Phi$ be the location of the target user on Earth's surface, then the receiver satellites are those within its local domes, i.e., $y \in \mathcal{B}$, where $\mathcal{B} = \cup_{m=1}^{M}\mathcal{B}_m$ and is the union of the local domes across all the shells in the constellation, are capable of receiving its signal. Moreover, we assume that all the ground users are transmitting at the same transmit power, denoted by ρ_{t}, and deploy the same type of antenna with gain pattern denoted by G_{t}. The uplink received signal vector $\mathbf{z} \in \mathbb{C}^{N\times 1}$ from those satellite branches, assuming that the satellites utilize an antenna with gain pattern denoted by G_{r}, is obtained as follows

$$\mathbf{z} = \mathbb{1}_{y_n\in\mathcal{B}}\left[\sum_{x_k\in\Phi\cap\mathcal{A}_n}\sqrt{\rho_{\mathrm{t}}G_{\mathrm{t}}(\boldsymbol{\varphi}_k)G_{\mathrm{r}}(\boldsymbol{\varphi}_k)}\mathbf{h}_k s_k + \mathbf{o}\right], \tag{10}$$

where $\mathbf{h}_k, \boldsymbol{\varphi}_k \in \mathbb{C}^{N\times 1}$ with subscript k which corresponds to the index of the ground user, s_k is the transmitted signal with mean 0 and variance 1, $\mathbf{o} \sim \mathcal{CN}(0, W_{\mathrm{s}}\mathbf{I}_N)$ and is an $N \times 1$ additive Gaussian noise vector where $\mathbf{I}_N$ is an $N \times N$ identity matrix, and $\mathbb{1}_{y_n\in\mathcal{B}}$ is a logical operator that is 1 for the satellites within the local dome and 0 for the rest.

We first evaluate the signal and interference at each satellite receiver, where the interference is generated from the ground users within the satellites' footprints which are located within the target user local dome, i.e., at $x_i \in \Phi \cap \mathcal{A}_n \backslash x_{\mathrm{o}}$, as illustrated in Fig. 3. Hence, we denote the signal transmitted by the target user as S_{o}. Accordingly, the received signal vector at the nth satellite within the target user local dome can be expanded from (10) as follows

$$z_{n,\mathrm{o}} = \sqrt{\rho_{\mathrm{t}}G_{\mathrm{t}}(\varphi_{n,\mathrm{o}})G_{\mathrm{r}}(\varphi_{n,\mathrm{o}})}h_{n,\mathrm{o}}s_{\mathrm{o}}$$

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$$+ \sum_{x_i \in \Phi \cap \mathcal{A}_n \setminus x_\mathrm{o}} \sqrt{\rho_\mathrm{t} G_\mathrm{t}(\varphi_{n,i}) G_\mathrm{r}(\varphi_{n,i})} h_{n,i} s_i + o. \quad (11)$$

Hence, the power at the nth satellite receiver orbiting in the mth shell from the target user located at x_o is calculated as

$$S_{n,\mathrm{o}} = \rho_\mathrm{t} G_\mathrm{t}(\varphi_{n,\mathrm{o}}) G_\mathrm{r}(\varphi_{n,\mathrm{o}}) l(\varphi_{n,\mathrm{o}}) \zeta(\varphi_{n,\mathrm{o}}). \quad (12)$$

Moreover, the aggregate interference power at the satellite receiver is the summation of all the active users that are within its footprint $\mathcal{A}_n$, given as follows

$$I_n \triangleq \sum_{x_i \in \mathcal{A}_n \setminus x_\mathrm{o}} \rho_\mathrm{t} G_\mathrm{t}(\varphi_{n,i}) G_\mathrm{r}(\varphi_{n,i}) l(\varphi_{n,i}) \zeta(\varphi_{n,i}). \quad (13)$$

Note that we assume that the antenna gains within the effective beamwidth are at their maximum, thus the angle subscript will be dropped. Furthermore, the SINR of the target ground user at the nth satellite receiver is formulated as follows

$$\mathsf{SINR}_\mathrm{o} = \frac{S_{n,\mathrm{o}}}{I_n + W_\mathrm{s}}. \quad (14)$$

On the other hand, the maximum achievable rate assuming Shannon's limit is defined as

$$\begin{aligned} \mathcal{P}(\mathcal{R}) &\triangleq \mathbb{P}\left(B \log_2(1 + \mathsf{SINR}) \geq \mathcal{R}\right) \\ &= \mathbb{P}\left(\mathsf{SINR} \geq \gamma_\mathrm{o}\right) = \mathcal{P}_\mathrm{c}(\gamma_\mathrm{o}), \end{aligned} \quad (15)$$

where $\mathcal{R}$ is the threshold achievable rate, B is the signal bandwidth, and $\gamma_\mathrm{o} \triangleq 2^{\mathcal{R}/B} - 1$ is the target threshold SINR. The main advantage of diversity and the utilization of a super node is the ability to maximize the obtained SINR and inherently the data rate. This can be achieved by deploying intelligent mechanisms that rely on reducing the effect of impairments while facilitating hand-over and spatial beam-forming. In this work, we address two main types of analysis; (i) satellite selection and (ii) satellite combining.

III. Coverage Probability and Rate

Due to the large swath of the satellite beam, the number of active ground users that transmit is relatively high [49]. Hence, in order to derive the coverage probability, we derive the generated interference within a typical satellite footprint. The aggregate interference at the satellite receiver is the summation of all active users located within the satellite footprint, $\mathcal{A}$.

Theorem 1: When considering the large footprints that satellites provide, the number of interfering devices converges to the average value of $\lambda||\mathcal{A}||$. Hence the variation caused by the spatial process across the interference become extremely small and the distribution converges into a normal distribution.

Since the variations in the interference are considered to be negligible, the average is sufficient.

Lemma 2: The average interference seen at any satellite per shell is derived as follows

$$\bar{I}_m = 2\pi\lambda R_\oplus^2 \rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} \int_0^{\varphi_m^{\max}} l_m(\varphi) \bar{\zeta}_m(\varphi) \sin\varphi \, \mathrm{d}\varphi. \quad (16)$$

where $\bar{\zeta}$ is the average small-scale path-loss power.

Proof: See Appendix B. ■

Remark 1: For channels that are dominated by LoS, where the small-scale fading is negligible, i.e., $\bar{\zeta} = 1$, the average interference is expressed in closed-form as follows

$$\begin{aligned} \bar{I}_m &= \pi\lambda\rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} \alpha_m \\ &\ln\left[\frac{R_\oplus^2 + (R_\oplus + R_m)^2 - 2R_\oplus(R_\oplus + R_m)\cos\varphi_\mathrm{m}^{\max}}{R_m^2}\right]. \end{aligned} \quad (17)$$

Note that since the satellites are assumed to have the same beamwidth, then their coverage footprint covers the same area. Therefore, the average interference experienced is identical for all the satellites that are within a shell and only varies for satellite that are located on different shells.

A. Satellite Selection

The network can select the most optimal channel conditions available by selecting to decode the signal coming from the satellite receiver providing the highest SINR. The maximum SINR at the super node can be mathematically obtained as follows

$$\mathsf{SINR}_m^{\mathrm{SS}} = \underset{y_n \in \mathcal{B}_m}{\operatorname{argmax}} \left[\frac{\rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} |\mathbf{h}_\mathrm{o}|^2}{\sum\limits_{x_i \in \mathcal{A}_n \setminus x_\mathrm{o}} \rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} |\mathbf{h}_i|^2 + W_\mathrm{s}}\right], \quad (18)$$

where $\mathbf{h}_\mathrm{o}, \mathbf{h}_i \in \mathbb{C}^{N_m \times 1}$. Accordingly, the probability of coverage is obtained in the following lemma.

Lemma 3: The probability of coverage per shell utilizing this diversity model is the probability that the SINR for satellite selection is higher than the threshold is derived as follows

$$\begin{aligned} \mathcal{P}_m^{\mathrm{SS}}(\gamma_\mathrm{o}) = 1 - \exp\left(-\bar{N}_m\right) \sum_{n=0}^{\infty} \frac{1}{n!} \\ \left(\bar{N}_m \int_\varphi F_\zeta\left(\frac{\gamma_\mathrm{o}[\bar{I} + W_\mathrm{s}]}{\rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} l(\varphi)}\right) f_\mathcal{B}(\varphi) \mathrm{d}\varphi\right)^n, \end{aligned} \quad (19)$$

where $F_\zeta(\cdot)$ is the cumulative distribution function (CDF) of the excess path-loss and $f_\mathcal{B}(\varphi) = \sin\varphi/(1 - \cos\varphi_m^{\max})$ is the distribution of the satellites in the local dome. Moreover, the integral limits $\varphi \in [0, \varphi_m^{\max}]$.

Proof: See Appendix C. ■

Remark 2: If the most optimal choice is the nearest satellite, i.e., the contact satellite, then (19) simplifies to [43]

$$\mathcal{P}_m^{\mathrm{SS}}(\gamma_\mathrm{o}) = 1 - \int_\varphi F_\zeta\left(\frac{\gamma_\mathrm{o}[\bar{I}_m + W_\mathrm{s}]}{\rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} l(\varphi)}\right) f_\varphi(\varphi) \mathrm{d}\varphi, \quad (20)$$

where $f_\varphi(\varphi) = \frac{N_m}{2} \sin\varphi \exp\left(-\frac{N_m}{2}[1 - \cos\varphi]\right)$ which is the contact satellite distribution [34].

For a multi-shell network, the SINR for the selection scheme is the maximum SINR across all the satellites that are within all the local domes of the target ground user. Accordingly, the SINR is the maximum across all shells as follows

$$\begin{aligned} \mathsf{SINR}^{\mathrm{SS}} &= \underset{m \in M}{\operatorname{argmax}} \left[\mathsf{SINR}_m^{\mathrm{SS}}\right] \\ &= \underset{y_n \in \mathcal{B}}{\operatorname{argmax}} \left[\frac{\rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} |\mathbf{h}_\mathrm{o}|^2}{\sum\limits_{x_i \in \mathcal{A}_n \setminus x_\mathrm{o}} \rho_\mathrm{t} G_\mathrm{t} G_\mathrm{r} |\mathbf{h}_i|^2 + W_\mathrm{s}}\right], \end{aligned} \quad (21)$$

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where $\mathbf{h}_{\rm o}, \mathbf{h}_i \in \mathbb{C}^{N\times 1}$. Similarly, we obtain the probability of coverage for the selection scheme across a multi-shell network in the following lemma.

Theorem 2: The coverage probability for a multi-shell network that utilizes the selection diversity scheme is obtained as

$$\mathcal{P}^{\rm SS}(\gamma_{\rm o}) = 1 - \prod_{m=1}^{M}\sum_{n=0}^{\infty}\frac{\exp\left(-\bar{N}_m\right)}{n!} \left(\bar{N}_m \int_{\varphi} F_{\zeta}\left(\frac{\gamma_{\rm o}[\bar{I}_m + W_{\rm s}]}{\rho_{\rm t}G_{\rm t}G_{\rm r}l(\varphi)}\right) f_{\mathcal{B}}(\varphi){\rm d}\varphi\right)^n, \quad (22)$$

where the integral limits are $\varphi \in [0, \varphi_m^{\max}]$.

Proof: See Appendix D. ■

Note that each shell will have its own path-loss, number of satellites per shell, maximum Earth-centered angle, and average interference.

Remark 3: The nearest shell is typically the most optimal, however, this may change due to the small-scale fading, which may result in variations to the signal power.

B. Combining Diversity

Combining the signals coming from the different satellites within the local domes enables the system to exploit the advantages of spatial diversity. This is typically achieved by multiplying each link with different weights followed by summing the product at the super node to reconstruct the best possible received signal as follows

$$\hat{s}_{\rm o} = \mathbf{w}_{\rm o}^{\sf H}\mathbf{z}_{\rm o} = \sqrt{\rho_{\rm t}G_{\rm t}G_{\rm r}}\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_{\rm o}s_{\rm o} + \underbrace{\sqrt{\rho_{\rm t}G_{\rm t}G_{\rm r}}\sum_{x_i\in\mathcal{A}\setminus x_{\rm o}}\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_i s_i}_{\text{interference}} + \mathbf{w}_{\rm o}^{\sf H}\mathbf{o}, \quad (23)$$

where $\mathbf{w}_{\rm o} \in \mathbb{C}^{N_m\times 1}$ is the user-specific weight vector chosen to maximize the SINR for the target user located at $x_{\rm o}$. By exploiting the Gaussian signaling assumption, the user-specific SINR is derived as follows

$$\mathsf{SINR}_m^{\rm CD} = \frac{\rho_{\rm t}G_{\rm t}G_{\rm r}|\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_{\rm o}|^2}{\sum\limits_{x_i\in\mathcal{A}\setminus x_{\rm o}}\rho_{\rm t}G_{\rm t}G_{\rm r}|\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_i|^2 + W_{\rm s}\left\|\mathbf{w}_{\rm o}\right\|^2}. \quad (24)$$

The choice of the user-specific weights controls the quality of the reconstructed signal [50]. For instance, in noise-limited links, MRC is deployed where the weights are selected to be the conjugate of the channel state. In the presence of interference, however, MRC does not achieve the best outcome and an enhanced outcome can be achieved by deploying optical combining [50]. While both methods require prior knowledge of the iCSI at the receiver, optimal combining requires the knowledge of the interference correlation as well.

While iCSI can be theoretically achieved using preassigned pilots [51], the interference correlation is extremely difficult and often impossible to achieve, especially in rapidly changing channels such as LEO links. Moreover, due to the large footprint, the variation of the aggregate interference is negligible, as per Theorem 1. Hence, we assume that the combiner treats the interference similarly to white noise, which is typically assumed in diversity analysis using stochastic geometry for terrestrial networks [52], [53].

Lemma 4: The coverage probability of the combining diversity for a given shell can be approximated as follows

$$\mathcal{P}_m^{\rm CD}(\gamma_{\rm o}) = 1 - \frac{10^{L/3}}{\gamma_{\rm o}}\sum_{l=0}^{2L}(-1)^l\xi_l{\rm Re}\left\{\frac{\gamma_{\rm o}}{\beta_l}\mathcal{L}_m\left(\frac{\beta_l}{\gamma_{\rm o}}\right)\right\}, \quad (25)$$

where L is a predetermined integer, $\beta_l = \frac{L\ln 10}{3} + j\pi l$, $\xi_0 = \frac{1}{2}$, $\xi_1 = 1$ for $1 \le l \le L$, $\xi_{2L} = \frac{1}{2^L}$, $\xi_{2L-l} = \xi_{2L-l+1} + 2^{-L}\binom{L}{l}$ for $0 < l < L$. Moreover, $\mathcal{L}_m(s)$ is the Laplace transform of the sum of a homogeneous PPP derived in (45).

Proof: See Appendix E ■

For a multi-shell combining diversity network, the signals are combined from the satellite receivers across the different shells within the network. Similarly, this is achieved by multiplying each link with different weights followed by summing the product from the different shells at the CPU to reconstruct the best possible received signal as follows

$$\hat{s}_{\rm o} = \sum_{m=1}^{M}\mathbf{w}_{\rm o}^{\sf H}\mathbf{z}_{\rm o} = \sum_{m=1}^{M}\sqrt{\rho_{\rm t}G_{\rm t}G_{\rm r}}\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_{\rm o}s_{\rm o} + \underbrace{\sqrt{\rho_{\rm t}G_{\rm t}G_{\rm r}}\sum_{x_i\in\mathcal{A}\setminus x_{\rm o}}\sum_{m=1}^{M}\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_i s_i}_{\text{interference}} + \sum_{m=1}^{M}\mathbf{w}_{\rm o}^{\sf H}\mathbf{o}, \quad (26)$$

where the interference is only a function of the shell number and thus is independent of the shell summation. Similarly, by exploiting the Gaussian signaling assumption, the SINR of the multi-shell network using MRC combining diversity is obtained as follows

$$\mathsf{SINR}^{\rm CD} = \frac{\sum\limits_{m=1}^{M}\rho_{\rm t}G_{\rm t}G_{\rm r}|\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_{\rm o}|^2}{\sum\limits_{x_i\in\mathcal{A}_m\setminus x_{\rm o}}\rho_{\rm t}G_{\rm t}G_{\rm r}\sum\limits_{m=1}^{M}|\mathbf{w}_{\rm o}^{\sf H}\mathbf{h}_i|^2 + W_{\rm s}\sum\limits_{m=1}^{M}\left\|\mathbf{w}_{\rm o}\right\|^2}. \quad (27)$$

By utilizing the weights according to MRC, the coverage probability is calculated using the following lemma.

Theorem 3: The coverage probability of the combining diversity for a multi-shell network is approximated as follows

$$\mathcal{P}^{\rm CD}(\gamma_{\rm o}) = 1 - \frac{10^{L/3}}{x}\sum_{l=0}^{2L}(-1)^l\xi_l{\rm Re}\left\{\frac{x}{\beta_l}\prod_{m=1}^{M}\mathcal{L}_m\left(\frac{\beta_l}{x}\right)\right\}. \quad (28)$$

Proof: See Appendix F ■

IV. Simulation and Numerical Results

The Monte-Carlo simulator generates two independent random processes as follows; (i) a point process that represents the randomly generated satellite points on a sphere with multi-shelled altitudes $R_{\oplus} + R_m$, and (ii) a process for the ground users randomly scattered on Earth's surface. To achieve a uniform distribution of points on the spherical surface, the coordinates are generated as, $\theta \sim \mathcal{U}(0, 2\pi)$ and

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TABLE II
SIMULATION PARAMETERS

Parameter	Value	Unit
$R_\oplus$	6731	km
f_c	2	GHz
β	0.4	
$\mu_{\text{LoS}}, \mu_{\text{nLoS}}, \sigma_{\text{LoS}}, \sigma_{\text{nLoS}}$	0.4, 0, 1, 5, 2	dB
ρ_t	10	dBm
G_t, G_r	3, 2	dBi
κ	-20	dB
ψ_o	2π (isotropic)	radians
W_s	-160	dB

$\phi \sim \text{asin}[\mathcal{U}(-1,1)]$ where $\mathcal{U}(a,b)$ is the uniform random variable between a and b.

Table II summarizes the parameters used in the simulation. The carrier frequency is set at 2 GHz and we follow the channel model parameters obtained based on real measurements in [54]. At these low frequencies, we can assume that the air absorption attenuation is negligible based on the International Telecommunication Union (ITU) model [55]. We assume that the shadowed small-scale fading magnitude follows a log-normal distribution, modeled in [54], as follows (29), as shown at the bottom of the page,where p_{LoS} is the probability of LoS and its complement, p_{nLos} is the probability of non-line-of-sight (NLoS). The LoS probability is formulated as follows

$$p_{\text{LoS}}(\varphi) = \exp\left(-\frac{\beta \sin\varphi}{\cos\varphi - \alpha_m}\right), \tag{30}$$

where $\beta, \mu_{\text{LOS}}, \sigma_{\text{LoS}}, \mu_{\text{nLOS}}$, and σ_{nLoS} depend on the propagation environment [54]. Hence, the CDF of the small-scale fading is obtained as follows

$$F_\zeta(x) = \frac{1}{2} + \frac{p_{\text{LoS}}}{2}\text{erf}\left[\frac{10\log_{10} x + \mu_{\text{LOS}}}{\sqrt{2}\sigma_{\text{LoS}}}\right] + \frac{p_{\text{nLos}}}{2}\text{erf}\left[\frac{10\log_{10} x + \mu_{\text{nLOS}}}{\sqrt{2}\sigma_{\text{nLoS}}}\right], \tag{31}$$

where the probability density function (PDF) is given as

$$f_\zeta(x) = \frac{p_{\text{LoS}}}{x\sigma_{\text{LoS}}\sqrt{2\pi}}\exp\left[-\frac{(10\log_{10} x + \mu_{\text{LOS}})^2}{2\sigma_{\text{LoS}}^2}\right] + \frac{p_{\text{nLos}}}{x\sigma_{\text{nLoS}}\sqrt{2\pi}}\exp\left[-\frac{(10\log_{10} x + \mu_{\text{nLOS}})^2}{2\sigma_{\text{nLoS}}^2}\right]. \tag{32}$$

Note that other small-scale fading distribution models can be used in the equations.

Fig. 4 validates the equations derived for the two diversity models for both the single-shell and multi-shell architectures. It demonstrates a close fit between the Monte-Carlo simulations and the theoretical equations obtained in the previous section for the coverage probability. The simulations are carried out using a ground user point process of $\lambda_o = 1$ user per 100 km^2 and a duty cycle of D_o of 25%. The single-shell architecture utilizes $N_m = 900$ at $R_m = 600$ km,

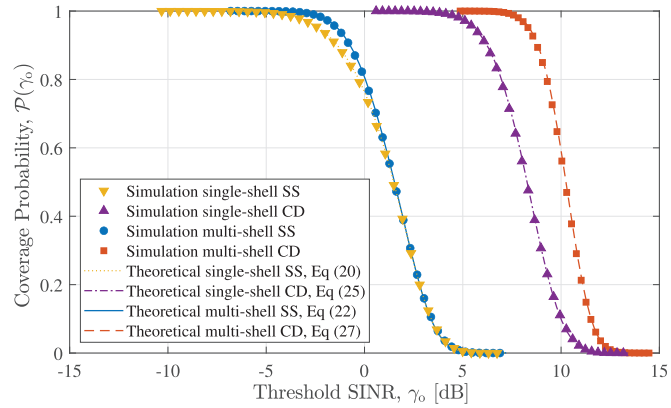

Fig. 4. Coverage probability comparison between simulation and the derived closed form equations for single-shell and multi-shell architectures. Theoretical formulations show an exact fit with the simulations.

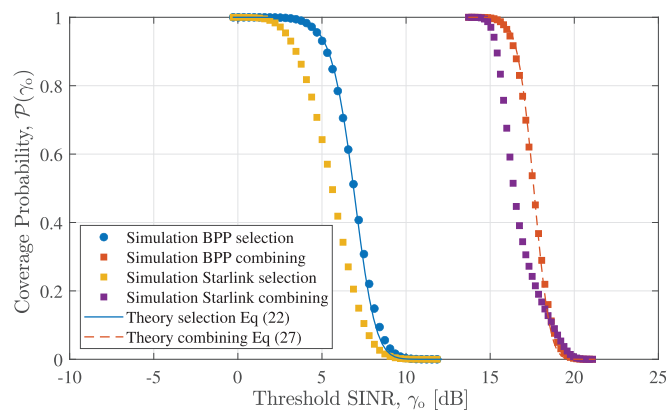

Fig. 5. Comparison between Starlink's Phase 2 three-shelled network and the theoretical models provided in this work for both diversity techniques.

whereas the multi-shell point process corresponds to a three-shell architecture with $N_m = \{900, 400, 100\}$ at $R_m = \{600, 900, 1200\}$ km. It is clear that the MRC combining diversity provides an enhancement in the coverage probability for both the single- and multi-shell architectures. Moreover, the multi-shell architecture provides further enhancement of coverage due to the increased number of receivers on the different shells for the MRC combining diversity. However, not much enhancement can be noticed in the satellite selection technique. This can be attributed to the fact that the nearest satellite is typically the selected receiver and the extra shells did not provide much enhancement except at much lower threshold.

Next, we depict the performance of a practical network and compare its performance with the model provided in this work. We utilize Starlink's Phase 2 architecture projected to be launched in 2027 [56] which is based on a three-shell satellite network with $N_m = \{2493, 2478, 2547\}$ at $R_m = \{335.9, 340.8, 345.6\}$ km at an inclination of $\{42°, 48°, 53°\}$. Although Starlink's performance is intractable due to the irregularity of the architecture, the theoretical models in

$$\zeta[\text{dB}] \sim p_{\text{LoS}}(\varphi)\mathcal{N}\left(-\mu_{\text{LoS}}, \sigma_{\text{LoS}}^2\right) + p_{\text{nLoS}}(\varphi)\mathcal{N}\left(-\mu_{\text{nLoS}}, \sigma_{\text{nLoS}}^2\right), \tag{29}$$

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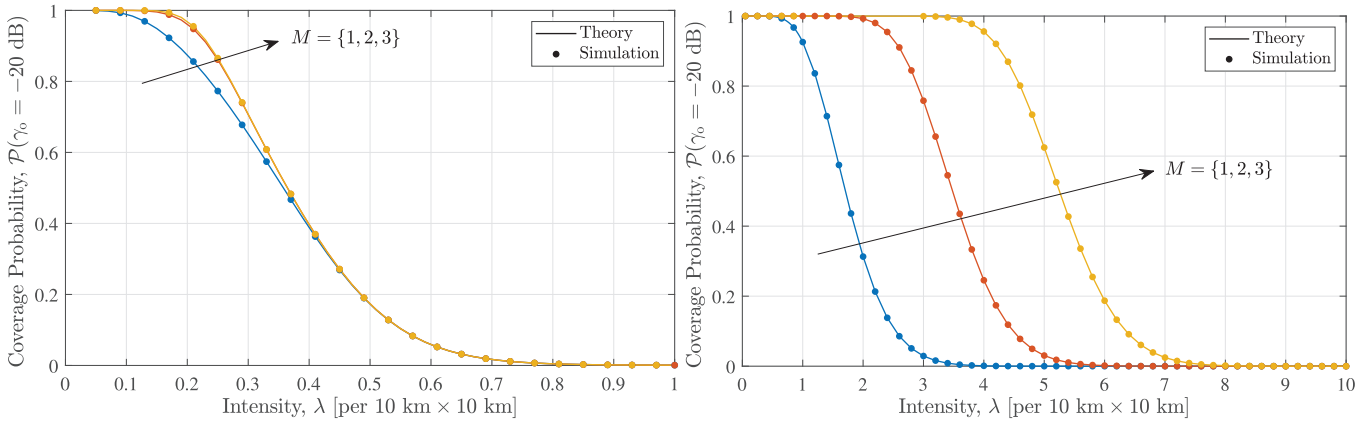

(a) Satellite selection (b) Combing diversity

Fig. 6. The effect of ground user intensity on the coverage probability assuming $\gamma_o = -20$ dB in a multi-shelled architecture for (a) satellite selection and (b) combining diversity.

this work provide a close fit for both the satellite selection and the combining scheme with slight differences as Fig. 5 illustrates with the assumption of a ground user intensity of $\lambda_o = 1$ user per 100 km^2 and a duty cycle of D_o of 25% for the same number of shells, altitudes, and satellites per shell.

In order to understand the effect of the ground user intensity, we set the threshold SINR to $\gamma_o = -20$ dB, and sweep the intensity for both diversity models. Fig. 6 depicts the coverage probability as a function of the ground user intensity for a 900 single-shell at 600 km (in blue), two-shells each with 900 satellites at 600 and 900 km respectively (red), and three-shells each with 900 satellites at 600, 900, and 1200 km respectively (yellow). Fig. 6a illustrates the performance of the satellite selection scheme where increasing the number of shells enhances the coverage probability slightly but saturates really quickly. Similar to the previous analysis, this is due to the nearest satellite being the most dominant receiver and thus regardless of the number of shells, the coverage probability does not enhance significantly. Fig. 6b depicts a contrary scenario where more shells in the combining diversity scheme enhance the coverage probability significantly and enable more ground users to utilize the network. This is mainly due to more satellites being able to receive the signal regardless of their altitude. As a result, we can conclude that combining diversity is significantly enhanced as the number of shells increase.

Moreover, we analyze the effect of the number of satellites per shell on the coverage probability. In this analysis, we consider a two-shell network with the first and second shells at altitudes 600 km and 1200 km, respectively. Then, we sweep the number of the satellites within the two shells and analyze their effect on the coverage probability. In Fig. 7, we present the contour of the coverage probability as a function of the number of satellites within each shell. As illustrated, the coverage probability increases drastically, especially at the higher altitude shells. This is expected since the number of receivers is increasing drastically for the same number of ground users which results in higher coverage and connectivity without leading to higher interference, especially at higher altitude where the interference total is smaller relevant to lower

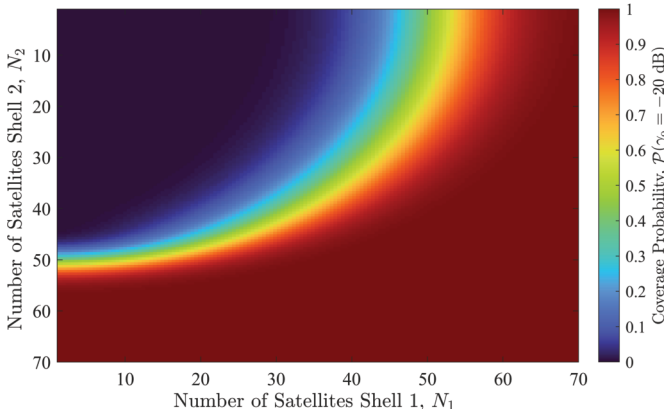

Fig. 7. Contour plot of the coverage probability as a function of the number of satellites in each shell. Shell 1 is the x-axis and shell 2 is the y-axis.

shells. In practice, nevertheless, this results in higher network costs and also more complex management schemes to maintain real diversity between those receivers.

As a result, the theoretical models derived can be utilized to provide an approximation of the performance of practical satellite networks when diversity techniques are deployed. Moreover, these theoretical models can be altered to account for any small-scale shadowing model or attenuation due to weather, since they are typically featured as a function of the elevation angle [57] and by extension the global zenith angle.

V. Conclusion

Next-generation satellite constellations are expected to utilize ISLs to enable communication diversity. However, current simulation techniques are extremely complex and require long running times. Hence, this article provides an analytical framework that derives the coverage probability for uplink in next-generation networks that deploy diversity. Monte-Carlo simulations show a close fit to the derived analytical results which can also provide a baseline to the performance of practical networks employing multi-shell architectures such

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as SpaceX's Starlink. The analysis provides insights into the effect of changing the network architecture which is a valuable resource to network designers as it allows them to depict the performance of the network prior to deployment. This work provides the pathway to address diversity in mega-satellite constellations by utilizing stochastic geometry and will be further extended to analyze more complicated diversity techniques in the future such as optimal combining.

APPENDIX

A. Proof for Channel Vector Lemma

The distance between a satellite and a ground user, where the global zenith angle between them is φ is expressed using the cosine rule as follows

$$d(\varphi) = \sqrt{R_{\oplus}^2 + (R_{\oplus} + R_m)^2 - 2R_{\oplus}(R_{\oplus} + R_m)\cos\varphi}. \tag{33}$$

Therefore, the complex channel gain can be expressed in phasor form as follows

$$h_{n,k} = \frac{4\pi f_{\mathrm{c}}}{c} d(\varphi)|g_{n,k}| \exp\left(-j\frac{2\pi f_{\mathrm{c}}}{c} d(\varphi) + \phi_g\right), \tag{34}$$

where ϕ_g is the phase difference due to small-scale channel fading. By utilizing the free space path-loss (FSPL) equation provided in (9), the received signal can be simplified as follows

$$h_{n,k} = \sqrt{l(\varphi)}|g_{n,k}| \exp\left(-j\frac{1}{2\sqrt{l(\varphi)}} + \phi_g\right). \tag{35}$$

By bringing back the form to baseband and reducing the small-scale fading to $g = |g|e^{j\phi_g}$, then this completes the proof.

B. Proof for Average Interference Lemma

We assume that the ground users are within a circular strip area on Earth's spherical surface right underneath the satellite, $f_{\mathcal{A}}(\varphi) = 2\pi R_{\oplus}^2 \sin\varphi$. Accordingly, the average interference is obtained by calculating the expectation over the aggregate interference as follows

$$\begin{aligned}\bar{I}_m &= \mathbb{E}_{\Phi}\mathbb{E}_{\zeta}\left[\sum_{x_i\in\mathcal{A}\setminus x_{\mathrm{o}}} \rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi_i)\zeta(\varphi_i)\right] \\ &\overset{(a)}{=} \mathbb{E}_{\Phi}\left[\sum_{x_i\in\mathcal{A}\setminus x_{\mathrm{o}}} \rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi_i)\bar{\zeta}(\varphi_i)\right] \\ &\overset{(b)}{=} \lambda\rho_{\mathrm{t}} \int_0^{\varphi_{\mathrm{m}}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi)\bar{\zeta}(\varphi) f_{\mathcal{A}}(\varphi)\ \mathrm{d}\varphi\ ,\end{aligned} \tag{36}$$

where (a) stems from the fact that ζ is independent from the users and thus its expectation can be taken inside the summation and (b) stems from invoking Campbell's theorem of sums for a stationary point process given as follows [58]

$$\mathbb{E}_{\Phi}\left[\sum_{x\in\Phi} f(x)\right] = \lambda\int_{\mathbb{R}^d} f(x)\ \mathrm{d}x. \tag{37}$$

C. Proof for Dominant Receiver Single-Shell Coverage Probability Lemma

The coverage probability of the dominant satellite receiver per shell is the complementary of the product of outage probability as follows [59]

$$\begin{aligned}\mathcal{P}_m^{\mathrm{SS}}(\gamma_{\mathrm{o}}) &= \mathbb{P}(\mathsf{SINR}_m^{\mathrm{SS}} \geq \gamma_{\mathrm{o}}) \\ &= 1 - \mathbb{P}(\mathsf{SINR}_m^{\mathrm{SS}} < \gamma_{\mathrm{o}}), \\ &= 1 - \mathbb{E}_N\left[\mathbb{P}(\mathsf{SINR}_m < \gamma_{\mathrm{o}})^n\right],\end{aligned} \tag{38}$$

where N is the a random variable representing the number of satellites within the target user local dome and n is the index of the satellite within this dome. Next we derive the probability of success at an individual satellite as follows

$$\begin{aligned}&\mathbb{P}(\mathsf{SINR}_m < \gamma_{\mathrm{o}}) \\ &= \mathbb{P}\left(\frac{S_{n,\mathrm{o}}}{I_n + W_{\mathrm{s}}} < \gamma_{\mathrm{o}}\right) \\ &= \mathbb{P}\left(\frac{\rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\sum\limits_{x_i\in\mathcal{A}_n\setminus x_{\mathrm{o}}} \rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi_{n,i})\zeta(\varphi_{n,i}) + W_{\mathrm{s}}} < \gamma_{\mathrm{o}}\right) \\ &\overset{(a)}{\approx} \mathbb{P}\left(\frac{\rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m + W_{\mathrm{s}}} < \gamma_{\mathrm{o}}\right) \\ &= \mathbb{P}\left(\zeta(\varphi_k) < \frac{\gamma_{\mathrm{o}}[\bar{I}_m + W_{\mathrm{s}}]}{\rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi_k)}\right) \\ &\overset{(b)}{=} \int_{\varphi} F_{\zeta}\left(\frac{\gamma_{\mathrm{o}}[\bar{I}_m + W_{\mathrm{s}}]}{\rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi)}\right) f_{\mathcal{B}}(\varphi)\mathrm{d}\varphi,\end{aligned} \tag{39}$$

where (a) stems from the approximation of the interference to its average due to the large footprint (refer to Theorem 1) and (b) from the definition of the CDF and by taking the expectation over the zenith angle. By taking the expectation over the number of satellites that are within the target user local dome we achieve the coverage probability as follows

$$\mathcal{P}_m^{\mathrm{SS}}(\gamma_{\mathrm{o}}) = 1 - \mathbb{E}_N\left[\left(\int_{\varphi} F_{\zeta}\left(\frac{\gamma_{\mathrm{o}}[\bar{I}_m + W_{\mathrm{s}}]}{\rho_{\mathrm{t}} G_{\mathrm{t}} G_{\mathrm{r}} l(\varphi)}\right) f_{\mathcal{B}}(\varphi)\mathrm{d}\varphi\right)^n\right], \tag{40}$$

where N follows a PPP distribution (4).

D. Proof for Dominant Receiver Multi-Shell Coverage Probability Theorem

The coverage probability of the multi-shell dominant satellite receiver is the complementary of the product of outage probability of all satellites across all shells as follows

$$\begin{aligned}\mathcal{P}^{\mathrm{SS}}(\gamma_{\mathrm{o}}) &= \mathbb{P}\left(\mathsf{SINR}^{\mathrm{SS}} \geq \gamma_{\mathrm{o}}\right) \\ &= 1 - \mathbb{P}\left(\mathsf{SINR}^{\mathrm{SS}} < \gamma_{\mathrm{o}}\right) \\ &= 1 - \prod_{m=1}^{M} \mathbb{P}\left(\mathsf{SINR}_m^{\mathrm{SS}} < \gamma_{\mathrm{o}}\right) \\ &= 1 - \prod_{m=1}^{M}\left[1 - \mathcal{P}_m^{\mathrm{SS}}(\gamma_{\mathrm{o}})\right].\end{aligned} \tag{41}$$

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E. Proof for Single-Shell Combining Diversity Lemma

Since the weights are assigned to be the conjugate of the channel coefficients through the iCSI, we can invoke the Cauchy-Schwarz inequality [50] as follows

$$\left|\mathbf{w}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}\right|^2 \leq \sum_{y_n\in\mathcal{B}_m}|w_{n,\mathrm{o}}|^2 \sum_{y_n\in\mathcal{B}_m}|h_{n,\mathrm{o}}|^2, \tag{42}$$

where the expression turns into an equality when the weights are equal to the channel coefficients. As a result, we can obtain the probability coverage as follows

$$\begin{aligned}\mathcal{P}_m^{\mathrm{CD}}(\gamma_{\mathrm{o}}) &= \mathbb{P}\left(\frac{\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}{\sum\limits_{x_i\in\mathcal{A}\setminus x_{\mathrm{o}}}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\frac{\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_i\mathbf{h}_i^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}{\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}+W_{\mathrm{s}}}\geq\gamma_{\mathrm{o}}\right)\\
&\overset{(a)}{=}\mathbb{P}\left(\frac{\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}{\sum\limits_{x_i\in\mathcal{A}\setminus x_{\mathrm{o}}}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\mathbf{h}_i^{\mathsf{H}}\mathbf{h}_i+W_{\mathrm{s}}}\geq\gamma_{\mathrm{o}}\right)\\
&\overset{(b)}{\approx}\mathbb{P}\left(\frac{\sum\limits_{y_n\in\mathcal{B}_m}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\geq\gamma_{\mathrm{o}}\right)\\
&\overset{(c)}{=}1-F_m(\gamma_{\mathrm{o}}),\end{aligned}\tag{43}$$

where (a) stems from the fact that the interfering channel coefficients $\mathbf{h}_i$ are independent of the weight vectors $\mathbf{h}_{\mathrm{o}}$ [53], (b) is due to the large number of interfering ground users, thus we replace the interference random variable with its mean (refer to Theorem 1) and (c) stems from the CDF definition, where $F_m(\cdot)$ is the CDF of the sum of the received signals obtained using the inverse Laplace transform of the Laplace transform of the random variable as follows

$$\begin{aligned}F_m(x) &= \mathcal{L}^{-1}\left[\frac{1}{s}\mathcal{L}_m(s);x\right]\\
&\overset{(a)}{=}\frac{10^{L/3}}{x}\sum_{l=0}^{2L}(-1)^l\xi_l\mathrm{Re}\left\{\frac{x}{\beta_l}\mathcal{L}_m\left(\frac{\beta_l}{x}\right)\right\},\end{aligned}\tag{44}$$

where (a) stems from Euler's inversion formula [60] and $\mathcal{L}_m(s)$ is the Laplace transform of the sum of a homogeneous PPP derived in (45), as shown at the bottom of the page,

where $f_\zeta(\cdot)$ is the distribution of the small-scale fading path-loss. Moreover, (a) is due to the independence of the small-scale path-loss and the distribution of the satellites and (b) is due to the probability generating functional of the PPP within the local domes whereby [58]

$$\mathbb{E}_N\left[\prod_{x\in N}v(\varphi)\right]=\exp\left(-\frac{N}{2}\int_{\varphi}[1-v(\varphi)]\sin\varphi\right). \tag{46}$$

F. Proof for Multi-Shell Combining Diversity Theorem

Similar to the single-shell scenario, we assume that the weights are assigned to the conjugate of the channel coefficients. Hence, we can obtain the probability coverage for a multi-shell system as follows

$$\begin{aligned}\mathcal{P}^{\mathrm{CD}}(\gamma_{\mathrm{o}}) &= \mathbb{P}\left(\mathsf{SINR}^{\mathrm{CD}}\geq\gamma_{\mathrm{o}}\right)\\
&=\mathbb{P}\left(\frac{\sum\limits_{m=1}^{M}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}{\sum\limits_{x_i\in\mathcal{A}\setminus x_{\mathrm{o}}}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\sum\limits_{m=1}^{M}\frac{\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_i\mathbf{h}_i^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}{\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}+W_{\mathrm{s}}}\geq\gamma_{\mathrm{o}}\right)\\
&\overset{(a)}{=}\mathbb{P}\left(\frac{\sum\limits_{m=1}^{M}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\mathbf{h}_{\mathrm{o}}^{\mathsf{H}}\mathbf{h}_{\mathrm{o}}}{\sum\limits_{x_i\in\mathcal{A}\setminus x_{\mathrm{o}}}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}\mathbf{h}_i^{\mathsf{H}}\mathbf{h}_i+W_{\mathrm{s}}}\geq\gamma_{\mathrm{o}}\right)\\
&\approx\mathbb{P}\left(\frac{\sum\limits_{m=1}^{M}\sum\limits_{y_n\in\mathcal{B}_m}\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\geq\gamma_{\mathrm{o}}\right)\\
&\overset{(b)}{=}1-F_{\Sigma}(\gamma_{\mathrm{o}}),\end{aligned}\tag{47}$$

where (a) stems from the fact that the interfering channel coefficients $\mathbf{h}_i$ are independent of the weight vectors $\mathbf{h}_{\mathrm{o}}$ [53] and that the interference is only conditioned on the shell index and is independent of the shell summation, and (b) stems from the CDF definition, where $F_\Sigma(\cdot)$ is the CDF of the double sum obtained using the inverse Laplace transform as follows

$$F_\Sigma(x)=\frac{10^{L/3}}{x}\sum_{l=0}^{2L}(-1)^l\xi_l\mathrm{Re}\left\{\frac{x}{\beta_l}\mathcal{L}_\Sigma\left(\frac{\beta_l}{x}\right)\right\}, \tag{48}$$

$$\begin{aligned}\mathcal{L}_m(s) &= \mathbb{E}\left[\exp\left(-s\sum_{y_n\in\mathcal{B}_n}\frac{\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\right)\right]\\
&\overset{(a)}{=}\mathbb{E}_N\left[\prod_{y_n\in\mathcal{B}_n}\mathbb{E}_\zeta\left[\exp\left(-\frac{s\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\right)\right]\right]\\
&\overset{(b)}{=}\exp\left(-\frac{N_m}{2}\int_0^{\varphi_m^{\max}}\left(1-\mathbb{E}_\zeta\left[\exp\left(-\frac{s\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi)\zeta(\varphi)}{\bar{I}_m+W_{\mathrm{s}}}\right)\right]\right)\sin\varphi\mathrm{d}\varphi\right)\\
&=\exp\left(-\frac{N_m}{2}\int_0^{\varphi_m^{\max}}\left[1-\int_0^{\infty}\exp\left(-\frac{s\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi)\zeta(\varphi)}{\bar{I}_m+W_{\mathrm{s}}}\right)f_\zeta(\zeta)\mathrm{d}\zeta\right]\sin\varphi\mathrm{d}\varphi\right),\end{aligned}\tag{45}$$

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where $\mathcal{L}_\Sigma(s)$ can be represented as the product of the single-shell Laplace transforms as follows

$$\begin{aligned}\mathcal{L}_\Sigma(s) &= \mathbb{E}\left[\exp\left(-s\sum_{m=1}^{M}\sum_{y_n\in\mathcal{B}_n}\frac{\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\right)\right]\\ &= \mathbb{E}\left[\prod_{m=1}^{M}\prod_{y_n\in\mathcal{B}_n}\exp\left(-\frac{s\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\right)\right]\\ &\overset{(a)}{=} \prod_{m=1}^{M}\mathbb{E}_N\left[\prod_{y_n\in\mathcal{B}_n}\mathbb{E}_\zeta\left[\exp\left(-\frac{s\rho_{\mathrm{t}}G_{\mathrm{t}}G_{\mathrm{r}}l(\varphi_{n,\mathrm{o}})\zeta(\varphi_{n,\mathrm{o}})}{\bar{I}_m+W_{\mathrm{s}}}\right)\right]\right]\\ &= \prod_{m=1}^{M}\mathcal{L}_m(s). \end{aligned} \tag{49}$$

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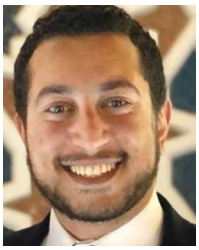

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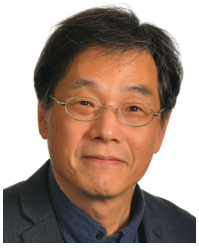

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